

Sneh Raj Sinha

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sneh30.github.io/portfolio

Summary

AI systems engineer and Electronics & Software Lead who takes projects end-to-end with cost and systems efficiency as key considerations: reduced LLM inference cost by up to 80–85% in PromptOS, built cost-optimization into DeepOps' remediation logic, and engineered embedded systems within 512KB RAM. Led software and electronics integration for a 50-member team's safety-critical competition vehicle. Graduating 2027.

Technical Skills

Languages: Python, Java, SQL

AI/Agent Systems: LangGraph, OpenAI Codex, Claude Code, OpenCode, OpenClaw, Hermes, Antigravity, OpenRouter, Ollama, llama.cpp

AI/ML: Hugging Face Transformers, LLM Inference, Multi-Agent Systems, Prompt Engineering, Edge AI

Backend/Cloud: FastAPI, PostgreSQL, Prometheus, REST APIs, AWS, Observability, Incident Response

Infra & Tooling: Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux, macOS, WASM, Metal (GPU)

Embedded/Hardware: ESP32, Raspberry Pi, Edge Impulse

Projects

Orchestra - Multi-Agent Research Orchestrator

LangGraph, FastAPI, PostgreSQL, Prometheus

github.com/Sneh30/Orchestra

- Built a multi-agent research system with LangGraph and FastAPI, backed by PostgreSQL and instrumented with Prometheus for observability
- Shipped a full test suite with a deterministic CI mode and packaged the full stack with Docker Compose for reproducible, one-command deployment

DeepOps - Autonomous SRE Platform

Python, Multi-Agent Systems

github.com/Sneh30/deepops

- Built a multi-agent platform that takes natural-language incident input and produces root cause analysis and remediation plans, with cost-optimization logic built into the remediation planner
- Core pipeline complete; currently in hardware integration and testing toward end-to-end automated resolution without human intervention

PromptOS - Local-First Prompt Compiler

Rust, Swift, llama.cpp, WASM, Metal

github.com/Sneh30/promptos

- Built a prompt compiler that reduces LLM cost and latency before a prompt reaches the model – a Rust core with a custom lexer, parser, AST, and ten optimization passes across four compilation modes
- Reduced inference cost by up to 80–85% in Economy mode via multi-pass token compression (tested against frontier LLM pricing); shipped as a self-contained macOS .app with a native Swift UI and Metal-accelerated local inference

ESP32-CAM Vision Suite - Embedded Computer Vision

C++, ESP32, Edge Impulse

github.com/Sneh30/esp32-cam-vision

- Built three embedded CV projects using math-driven approaches (HSV quantization, centroid tracking) instead of neural inference, running within 512KB RAM
- Implemented an autonomous color-tracking robot (on-device RGB565→RGB888→HSV conversion, L298N motor control) plus an image-collection and MJPEG streaming server supporting 15+ camera board variants

Experience

Infrastructure Engineer (Independent Engagement)

Aug 2026, One-month engagement

- Recruited for a one-month on-site engagement to strengthen infrastructure operations for a 400+ node exam-facility network; reduced infrastructure incidents and average recovery time by ~40% versus the prior comparable period

Software & Electronics Lead - Team Stratos

Jul 2025 – May 2026

- Led software and electronics integration for a 50-member team's competition vehicle; developed and deployed its ABS (Anti-lock Braking System) firmware, a safety-critical subsystem, validated through track testing, SAE India scrutineering, and competition

Open Source

webcmd (agentrhq/webcmd) – Merged PRs for bug fixes and documentation on an open-source browser-automation toolkit for AI agents

Education

Bangalore Institute of Technology

B.E. in Robotics and Artificial Intelligence, GPA – 8.0/10

Bangalore, Karnataka

Aug 2023 – Aug 2027